# Harish G

**AI Agents Engineer | Autonomous Agents, Tool Calling, Guardrails, Agent Orchestration**

Bengaluru, India | +91 7892855850 | harishgreddy.work@gmail.com | LinkedIn | GitHub | harishg.com

## Profile

I build agents that run unattended, and I instrument them well enough to know when one should be switched off. At CortexCraft.ai that is a fleet of Gemini Live phone agents taking real calls in production; on my own projects it is a tool host that hands an LLM a registry of declared functions and refuses to print a number no tool produced. I have also swapped a live platform's speech provider and backend without dropping traffic.

## Core Skills

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| **Languages** | Python · TypeScript · JavaScript · Java · Kotlin · SQL |
| **Agents** | Autonomous Agents · Tool Calling · MCP-style Registries · Output Guardrails · RAG · Orchestration |
| **Models** | Gemini Live · Groq · Ollama (local open-weight) · SmolLM2 QLoRA · Hugging Face · Whisper |
| **Backend** | FastAPI · Node.js / Express · Spring Boot · Celery · WebSockets · REST APIs |
| **Data** | PostgreSQL · pgvector · MongoDB · Neo4j · Redis |
| **Platform** | Google Cloud (Gemini, STT/TTS) · Docker · GitLab CI · Prometheus · Twilio · Plivo · WhatsApp |

## Experience

**Software Developer, CortexCraft.ai** Jan 2026 – Present

- Built and maintain the voice-agent platform (589 of 626 commits on the SaaS platform, 67 of 75 on the core engine): Gemini Live phone agents that run calls end to end over Twilio and Plivo, with per-agent RAG grounding, generated prompts, transcripts and an external REST API other products drive instead of forking the stack.
- Design the tool-calling orchestration and guardrails that constrain what an agent may do. Tools are declared functions, extracted call data is coerced to its declared type before anything downstream reads it, and handing a call to a human is an explicit path, not a fallback.
- Own production reliability for the call pipeline, which is where most of the real work lives. Fixed outbound routing, added reconnect handling for dropped Gemini 1006 sockets, and traced a Docker TLS failure to NAT hairpinning rather than restarting past it.

**Freelance Web & App Developer (Self-Employed)** 2023 – 2026

- Scoped and delivered full-stack projects directly with non-technical clients: React and Vite front ends, Python and Node.js services, automation scripts, then the fixes and deployment support afterwards.

## Selected Projects

**AI Personal CFO and Talo** *Kotlin, Compose, Hilt, Room/SQLCipher, ML Kit, Gemini Live*

- Sole author, 159 commits, 34 Gradle modules, 199 test files. An MCP-style in-process tool host exposes twelve deterministic engines through a JSON registry carrying `inputSchema`, `readOnlyHint` and `destructiveHint`; chat reads through the registry and never writes.
- An output guardrail rewrites any number in model text that is absent from the tool result as `[unverified]`, on a written rule: numbers from math, words from AI. On-device ML Kit OCR and the SMS parser are both scored against frozen evaluation fixtures.

**Dental Clinic AI Assistant** *Python, Flask, PostgreSQL/pgvector, Gemini, WhatsApp Business API*

- An agent that books appointments on its own: Gemini function calling with preconditions encoded in the tool descriptions, behind a role-based tool router admitting one tool per turn, over a 19-state WhatsApp state machine that hands off to a phone agent with context preserved.
- Chunked document RAG (800-token chunks, 100 overlap) embedded with `text-embedding-004` into pgvector, retrieved by cosine distance with a Postgres full-text fallback and role-scoped row filtering inside the retriever, so authorisation cannot be prompted around.

**KisanVoice AI** *Python, FastAPI, React, MongoDB, Google STT/TTS, Prometheus, Jenkins*

- 172 of 192 commits. Swapped the speech provider for Google Cloud STT/TTS and replaced the Node backend with FastAPI without interrupting live surveys; 24 locales (10 Indian) now come from a config-driven registry, so adding one is a config row rather than a deploy.
- Measured it rather than assumed it: every request carries an `X-Request-ID` echoed back, every log line is one JSON object carrying that ID, each response is logged with its status and duration in milliseconds, and a Prometheus instrumentator exposes `/metrics`.

**JiraFlow Agent** *Python, React, Jira API, GitHub API, LLM Agents*

- The place an agent earned in our own delivery process: it reads a Jira ticket, applies the change, and opens a pull request from an `ai/scrum-<id>` branch whose commit cites the ticket. The agent proposes and a human merges, so a wrong edit costs a rejected PR rather than a broken main branch.

## Education & Ways of Working

**B.E., Information Science and Engineering**, Vivekananda Institute of Technology, Bangalore

Feature branches, conventional commits, and 132 pytest files as the GitLab CI merge gate. I write the deployment runbooks and rollback procedures too.